Advanced Optical Fiber Communications

Problem 20: Transmitters, Solution

a) For OOK it is sufficient to turn the power on and off. For example, the lower EAM can be in off state and the data signal is then fed to the upper EAM. BPSK signals have either 0 or T phase, and to generate this both of the EAMs need to be switched. This is done by making sure that the upper EAM is in the on state when the lower is in the off state and vice versa.

b) The simplest solution is to set up the phases to be 0°, 90°, 180°, 270°. Then, electrical signals are generating from the data in such a way that only one EAM is in the on state at every given instant.

Problem 21: Long Range Transmission, Solution

a) With 10 sections the total accumulated CD in the link is 17000 ps/nm, and the number of required filter taps is 4 x 17000/1000 = 68.

b) The residual dispersion is 17000 - 10 x 15 x 100 = 2000 ps/nm, and the number of required filter taps is 4 x 2000/1000 = 8.

c) The impact from CD has a quadratic dependence on the symbol rate (pulse width), meaning that the number of taps required to compensate for a CD of 1000 ps/nm is 4 x (28/10)2 = 31.4, which we round upwards to 32.